# CAN Specification

VFDSXVEEP0047

# Complete Vehicle Specification

Specification Book

**Scope**
Applicable Document of Part 1

## Common Electrical Specification for Supplier
## **CES CAN**

**Range of validity**
Vinfast

Document Administrator: Nguyen Dinh Dung

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# CAN Specification

VFDSXVEEP0047

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| | | | |
|---|---|---|---|
| | Table 4, some abbreviations removed.<br>3.2 reduced wording.<br>3.3.1 reduced wording.<br>3.3.2.1 topology wording removed & diagram updated. Old table 3-1 removed. Section updated with revised wording. New diagram added for stubs near a ECU containing termination.<br>3.3.2.2 changed to describe connector pins.<br>3.3.2.3 changed to describe harness attributes.<br>3.3.2.4 Old table 3-2 not changed. Add further wire twist detail.<br>3.3.2.5 bus interface section wording revised. Addition of best practice rules. Add old section 3.3.2.6 here.<br>3.3.2.6 CAN split termination, revised wording.<br>3.3.2.7 CAN physical signalling, old figure 3-3 removed. Old table 3-5 updated.<br>3.3.2.8 Common mode choke, wording revised and diagrams added.<br>3.3.2.9 capacitors, wording revised.<br>3.3.2.10, ESD protection, wording revised and table added.<br>3.3.1.11 external ESD protection, wording revised.<br>3.3.1.12 Bus int. with wakeup func., wording revised.<br>3.3.1.13 controller oscillator, wording revised.<br>3.4.1 wording revised. Some of old 3.4.3 incorporated in here.<br>Old 3.4.2 message format removed.<br>Old 3.4.4 removed.<br>3.4.2 CAN transmission speed, wording revised. TSEG table added. Old figure 3-5 removed. Old table 3-8 removed.<br>Old 3.4.5 removed.<br>Old 3.4.6 removed.<br>3.4.3 bus off, wording revised.<br>Old 3.4.8 removed.<br>3.5 wording revised.<br>3.5.1 STmin stated.<br>Old 3.5.2, 3.5.2.1, 3.5.2.2, 3.5.2.3, 3.5.2.4, 3.5.2.4.1, 3.5.2.5, 3.5.2.5.1, 3.5.2.5.2, 3.5.2.5.3, 3.5.3, 3.5.4, 3.5.5, 3.5.6, 3.5.6.1 removed.<br>Old 3.6, 3.7 removed.<br>Old 3.8 now 3.6. Some wording revision.<br>Old NM_CL15ONOFF_001 removed. | | |

| | | | |
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| | | | |

# CAN Specification

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# 1. General Information

## 1.1 Language

This specification is written in English.

# 2. Introduction

## 2.1 Purpose of the document

The Controller Area Network (CAN) is a message-oriented Multi Master Bus System using CSMA/CR (Carrier Sense Multiple Access/Collision Resolution). For identification of separate messages, a unique Identifier (ID) with 11 Bit (according to CAN 2.0A), where the lowest ID represents the highest priority. Generally spoken, the CAN as a multi master system allows bus arbitration at all time for all ECU's with respect to their priority (message ID).

The [NS1] and [NS2] is based on the ISO/OSI model for layered data communication. This model outlines the seven distinct layers for general communication architecture in which the definitions of the physical layer of CAN are separated in two specifications. [NS2] specifies the requirements of high-speed CAN for transmission rates is 500 kbps. The specifications about the physical layer also contain the requirements to the physical media. In general, for the connection of all CAN nodes a two-wire differential bus is used.
The data link layer is described [NS1]. This specification defines the physical signalling of CAN as well as the associated mechanism of medium access control. In general, this corresponds to implementation of CAN-controller on MCUs.

## 2.2 Terms and definitions

This chapter provides the details about Terms and definitions.

### 2.2.1 Abbreviations

The following list of abbreviations are used throughout this document with the given meaning.

#### 2.2.1.1 Temperatures

| | |
|---|---|
| Tmin | Minimum operating temperature |
| Tamb | Ambient temperature |
| Tmax | Maximum operating temperature |
| TTest | Test temperature |

*Table 1 Temperatures*

#### 2.2.1.2 Voltage and Current

| | |
|---|---|
| UBmin | Lower operation voltage limit |
| UB | Operation voltage |
| UBmax | Upper operation voltage limit |
| Umax | Maximum voltage during test |
| Umin | Minimum voltage during test |
| UPP | Peak-peak voltage |
| Ueff | Root-mean-square value of the voltage |
| UTest | Testing voltage |
| IN | Nominal current |
| GND | Device ground |
| DUT | Device under test |
| UA, UT, US, UR | Voltage levels during cranking profile |

*Table 2 Voltage and current*

All specified voltages and currents are related directly to the DUT. Additional voltage drops at the energy harness are not included if not mentioned separately.

#### 2.2.1.3 Timing

| | |
|---|---|
| tr | Rise time |
| tf | Fall time |
| tTest | Test duration |

*Table 3 Timing*

#### 2.2.1.4 Other

| | |
|---|---|
| ACK | Acknowledgement |
| ACC | Clamp ACC or Radio |
| AUTOSAR | Automotive Open System Architecture |
| API | Application Program Interface |
| BR | Bit Rate |
| CAN | Controller Area Network |
| CCP | CAN Calibration Protocol |
| CL15 | Clamp 15 (Ignition state/line) equal to ignition=ON |
| CL30 | Clamp 30 |
| CL30S | Clamp 30, switched |
| CL50 | Clamp 50, switched, turning the starter on |
| CRC | Cyclic Redundancy Check |

| | |
|---|---|
| CSMA/CA | Carrier Sense Multiple Access/Collision Avoidance |
| CSMA/CR | Carrier Sense Multiple Access/Collision Resolution |
| DLC | Data Length Code |
| DTC | Diagnostic Trouble Code |
| ECU | Electronic Control Unit |
| EMC | Electromagnetic Compatibility |
| EOF | End of Frame |
| HS-CAN | High Speed CAN |
| IDE | Identifier Extension Flag |
| ID | Identifier |
| ISO | International Standardization Organization |
| LSB | Least Significant Bit |
| kbps | Kilo bits per second |
| MCU | Micro Controller Unit |
| MSB | Most Significant Bit |
| NM | Network Management |
| NA | Not Available |
| OBD | On-Board Diagnosis |
| OSI | Open System Interconnection |
| PDU | Protocol Data Unit |
| PLL | Phase Locked Loop |
| REC | Receive Error Count |
| RTR | Remote Transmission Request |
| ROM | Read-Only Memory |
| OEM | Original Equipment Manufacturer |
| SOF | Start of Frame |
| SP | Sample Point |
| SJW | Resynchronization Jump Width |
| $St_{min}$ | Minimum Separation Time |
| TBD/tbd | to be defined |
| TA | Target Address |
| TEC | Transmit Error Count |
| TTC | Time Triggered Communication |
| VCS_xxx | Vinfast Can Specification |

*Table 4 Abbreviations*

### 2.2.2 Definition

The following list of definitions are used throughout this document with the given meaning.

#### 2.2.2.1 Obligation of Requirement Wording

The following wording rules are used throughout this document.

**Shall**

The word "shall" is used to state a binding requirement. Such requirements are externally verifiable by manipulation or observation.

**Should**

The word "should" is used for preferences or desires in implementation that are not strictly binding.

**May**

The word "may" is used to denote options that are left to the implementer.

#### 2.2.2.2 Standard Tolerances

| | |
|---|---|
| Frequency | ±1% |
| Temperature | ±2°C |
| Humidity | ±5% |
| Times | +5%; -0% |
| Voltages | ±2% |
| Currents | ±2% |

*Table 5 Standards and tolerances*

#### 2.2.2.3 Standard Values

| | |
|---|---|
| Room temperature | TRT=23°C |
| Humidity | Frel=45%-75% RH |
| Test temperature | TTest = TRT |
| Nominal voltage | UN=12V |
| Operation voltage | UB=14V |

*Table 6 Standard values*

#### 2.2.2.4 Other terms

The following list of terms and definitions are used throughout this document with the given meaning.

**bit rate**

The number of bits transmitted per time, independent of bit representation.

**bit stuffing and de-stuffing**

Insertion and removal of fill-bits to provide bus state changes after a specific number of equal bits. This is required for periodic resynchronization when using a non-return-to-zero bit representation.

**bus**
A network topology in which all nodes are interconnected with each other using passive links that allow transmission in both directions [QD1].

**dominant and recessive**
The two states of the physical communication line are called dominant and recessive. The recessive state can be overwritten by the dominant state but not vice versa.

**frame**
A data unit used by the data link layer for transmission across the transfer medium with a specified position and meaning for each data bit [QD1].

**node**
An assembly that is connected to a communication network. Nodes are capable of communicating across the network following some communication protocol specification [QD1].

**non-return-to-zero**
Characteristic of binary signals where the signal level within a single bit time does not change and where no edges occur within a sequence of bits having equal signal levels.

**protocol**
A formal set of rules that define the information interchange between nodes [QD1]. **receiver**

A node is a receiver if it is not transmitting and if the bus is not idle [QD1].

**transmitter**
A node sending a frame onto the bus. It stays transmitter until the bus is idle again or until it has lost arbitration.

### 2.3 CAN and the OSI-reference Model

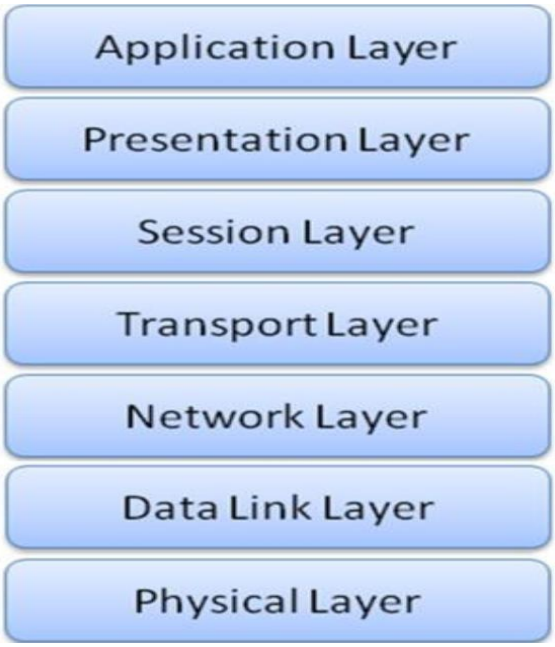



*Figure 1 OSI network model*

Figure OSI Network Model outlines the seven distinct layers for general communication architecture in which the layers shall be understand as described in section System description.
The aim of this document is to define all necessary requirements for the development of a CAN bus system used in the road vehicles.

## 3. System Description

### 3.1 Specific Terms and Definitions

The following list of terms and definitions are used throughout this document with the given meaning.

### 3.2 Overview

The Vinfast CAN communication system is to be implemented based on multiple ISO/Autosar standards.

### 3.3 Physical Layer

#### 3.3.1 Introduction

Particular physical layer attributes can be defined by the OEM. This is to ensure a robust and reliable CAN communication system that will last for the life of the vehicle.

### 3.3.2 High-Speed-CAN
### 3.3.2.1 BUS Topology

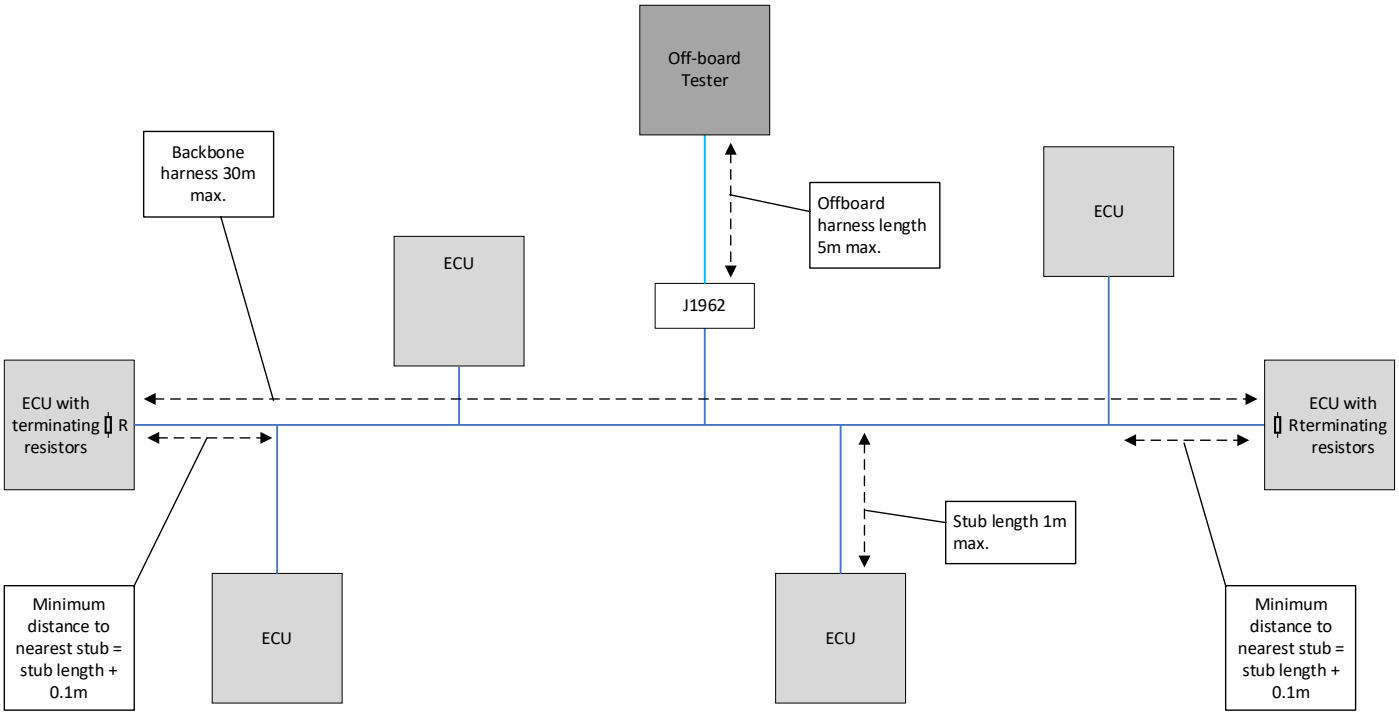



*Figure 2 Bus topology*

**Harness maximum lengths.**
The Backbone harness comprises of an un-shielded twisted wire pair that connects the two ECU which contain the terminating resistors.
The Stub harness comprises of an un-shielded twisted wire pair that is spliced into the Backbone harness and connects a single ECU to the Backbone.

**HARN01:**
Backbone harness: 30m max.
Stub harness: 1m max.
Off-board harness: 5m max.

Note: a stub harness shall not increase the overall length of the backbone harness.

**HARN02:**
**Maximum number of ECU per CAN bus.**
Maximum number: 21 ECU.
Note: If the CAN bus is connected to the J1962 (OBD) connecter, then the off-board tester counts as one ECU.
Option fit (after market) ECU. If connectors are fitted to a CAN bus to support option fit ECU but an ECU is not actually fitted at production, this is to be considered and counts as one ECU.

**HARN03:**
**Minimum distance to nearest stub for an ECU containing termination resistors.**
This applies to any stubs connected within 1.1m of a ECU containing termination resistors.
A stub harness shall not increase the overall length of the backbone harness. Given this the stub splice into the backbone harness shall be increased by 0.1m + the stub harness length.

Examples:

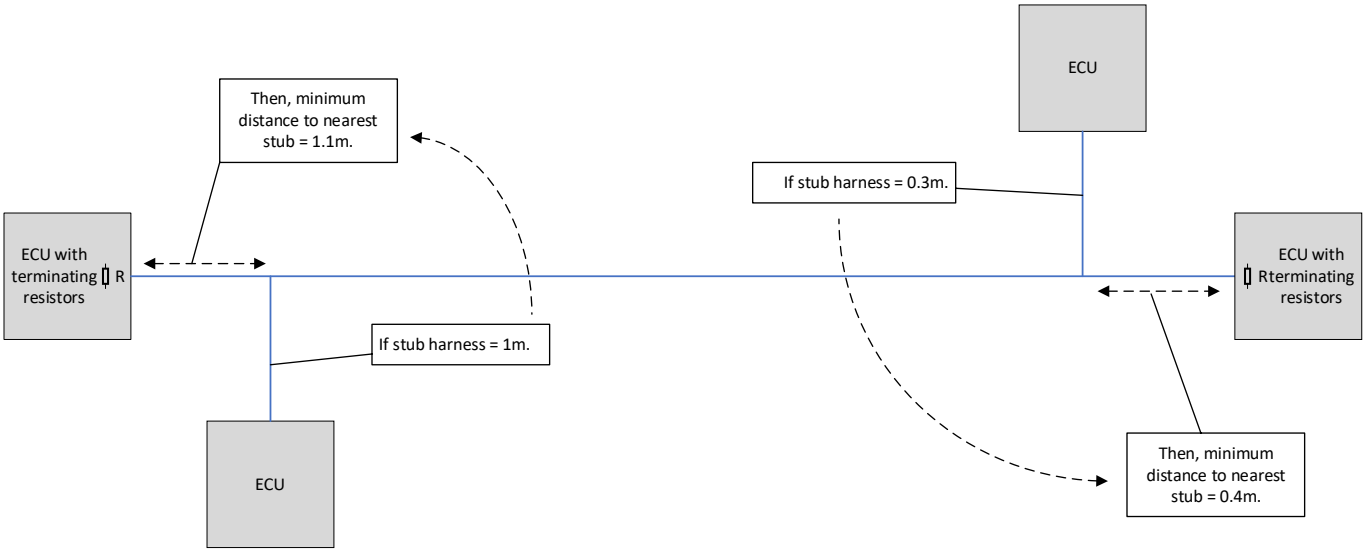



*Figure 3 Stub near termination ECU*

**HARN04:**
Two stubs cannot be originate from a single splice point on the backbone. This is not allowed on Vinfast application.

### 3.3.2.2 Harness/ECU CAN connector pins.

The CAN bus pins in the harness/ECU connectors shall be adjacent to each other.

In the diagram representation for the connector, the black pin represents one CAN bus wire. An adjacent pin could be any one of the blue pins.

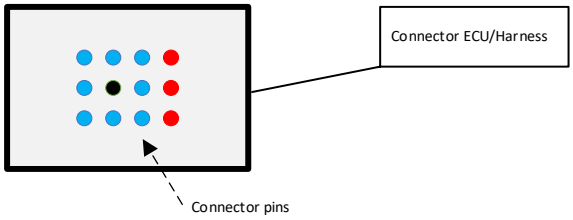



*Figure 4 Adjacent connector pins*

### 3.3.2.3 Harness considerations.

**HARN05:**
**Minimum distance between two stub connections on the backbone harness.**
On the main backbone harness stubs connections shall be no closer than: 0.1m.

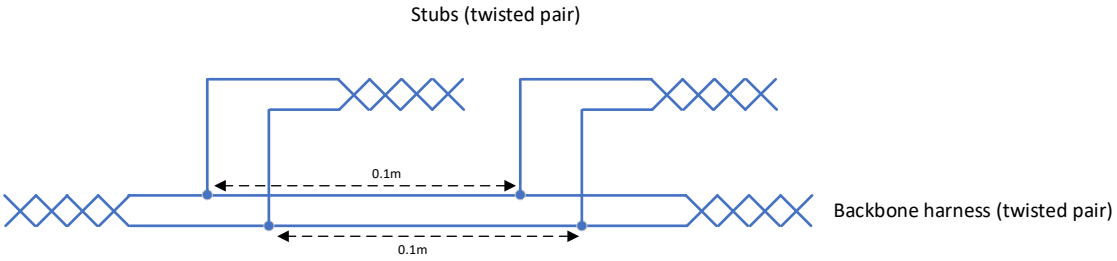



*Figure 5 Two stub connection on backbone*

**HARN06:**
**Stub connection to the backbone harness.**
When splicing a stub to the backbone harness, the physical connection points shall not be at the same location. This is to reduce the possibility of a bus short and reduce the cross-sectional area when incorporated into a vehicle harness.

Maximum backbone untwisted wire: 150mm.

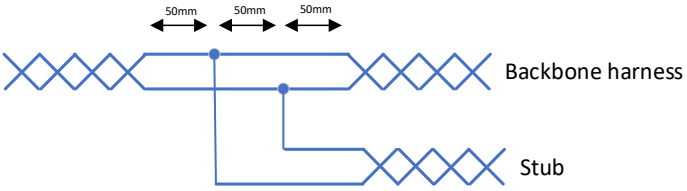



*Figure 6 Stub connection to backbone*

**HARN07:**
**Stub wire maximum untwist at the backbone/splice connection.**
Maximum stub untwisted wire: 100/50mm.

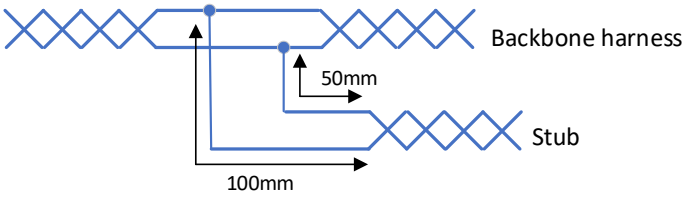



*Figure 7 Stub untwisted wire*

**HARN08:**
**Wire maximum untwist at a connector.**
Maximum at a connector untwisted wire: 50mm.

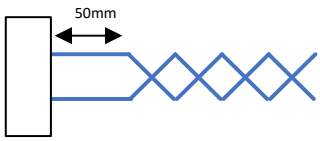



*Figure 8 Connector untwisted wire*

### 3.3.2.4 Wire attributes

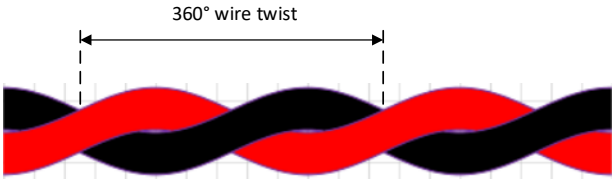
The wires used for HS-CAN shall have a wire gauge of: 0.35 mm²

For further specifications refer to [NS1] and [NS2].
The requirements of the wiring see table 7.

| Parameter | Symbol | Min | Norm | Max | Unit | Conditions |
|---|---|---|---|---|---|---|
| *Impedance* | Z | 90 | 120 | 140 | Ω | f=1 MHz |
| *Length Related Resistance* | $R_{Length}$ | | | 120 | mΩ/m | Single conductor |
| Specific Line Delay | $T_{Delay}$ | | | 5.5 | ns/m | Wire only |
| Rate of Twist | RateTwist | 33 | 40 | 50 | Twist/m | 360 degrees |

*Table 7 Physical Media Parameters for Unshielded Twisted Pairs*

Definition of one 360° wire twist.




The rate of twist shall be maintained throughout the length of the Backbone harness, the Stub harness and any external tester harness such as used for the off-board tester, (other than the given points of untwisted wire).

Any untwisted wire shall be kept to a minimum.

### 3.3.2.5 Bus interface HS_CAN

Figure 9 provides a recommended schematic that shows how the bus interface shall be implemented. The layout guidelines from the transceiver manufacturer shall be considered.

**HW01:**
It shall be ensured that the CAN transceiver is placed near the ECU connector due to EMC reasons and this is to be verified by PCB layout review.

**HW02:**
No other IC is to be placed between the connector and the CAN transceiver on the PCB.

**HW03:**
The PCB shall provide a ground plane under the CAN transceiver on the component layer.

**HW04:**
The PCB shall provide guard tracks either side of the CAN tracks between the edge connector and CAN transceiver. The guard tracks shall be on the same layer as the CAN tracks.

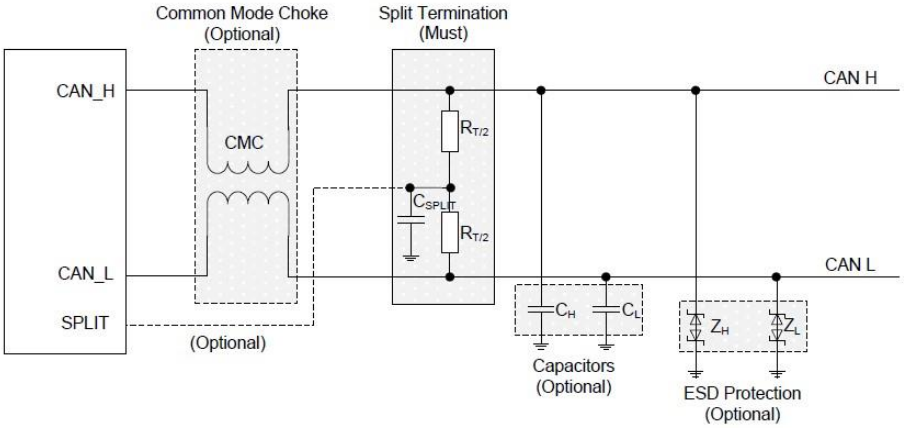



*Figure 9 High speed CAN bus interface*

### 3.3.2.6 High-speed-CAN Split Termination

**HW05:**
The PCB layout design shall include “footprint pads” at the appropriate location that allows termination resistors placement if the ECU is nominated as a termination node.
The ECUs that will populate the termination network will be defined specifically for each vehicle program as decided by the EEA team.
**Prior to sourcing any new ECU, this shall be checked by the commodity owner with the OEM level EEA team.**
Each CAN bus will support two modules hosting CAN termination in such a way that the CAN harness length is the longest between them, (the backbone harness).

**HW06:**
The termination resistors shall be metal film type and be a matched pair from the same batch. The tolerance between the resistor should be 1% and power dissipation of 0.3W (max value). See Fig 3-2 for the layout.

The termination components.

| Parameter | Unit | Min | Nom | Max | Comments |
|---|---|---|---|---|---|
| Resistor $R_{T/2}$ | Ω | 59 | 62 | 66 | |
| Capacitor $C_{SPLIT}$ | nF | | 4.7 | | |

*Table 8 Termination component*

### 3.3.2.7 High-speed-CAN physical signalling

| Parameter | Symbol | Unit | Value | | | Condition |
|---|---|---|---|---|---|---|
| | | | Min | Nom | Max | |
| Output voltage | $V_{CAN_H}$ | V | 2.0 | 2.5 | 3.0 | No Load |
| | $V_{CAN_L}$ | V | 2.0 | 2.5 | 3.0 | |
| Differential Output Bus Voltage | $V_{diff}$ | V | -0.5 | 0 | 0.5 | No Load |

*Table 9 Output Bus Voltage Parameter for Recessive state*

| Parameter | Symbol | Unit | Value | | | Condition |
|---|---|---|---|---|---|---|
| | | | Min | Nom | Max | |
| Output voltage | $V_{CAN_H}$ | V | 2.75 | 3.5 | 4.5 | Load RT/2 |
| | $V_{CAN_L}$ | V | 0.5 | 1.5 | 2.25 | |
| Differential Output Bus Voltage | $V_{diff}$ | V | 1.5 | 2.0 | 3.0 | Load RT/2 |

*Table 10 Output Bus Voltage Parameter for Dominant State*

### 3.3.2.8 Common mode choke

ECU may be subject to EMC disturbances on the vehicle platform. If the ECU is subject to EMC disturbances, then it may be required to include a common mode choke (option) into the design.

All ECU shall support the capability to fit a common mode choke on the PCB for the CAN bus. The PCB layout design shall include “footprint pads” at the appropriate location that allows a bus common mode choke placement should this be required.

**HW07:**
If an inductor is not required then the circuit is completed by using two zero ohm, (0Ω), resistors, inline on the CAN bus track, (Figure 10).

**HW08:**
If an inductor is required, then typically this shall be: 15 - 33µH, (Figure 11).

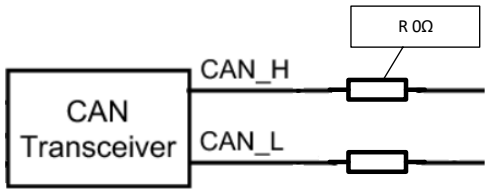



*Figure 10 Common mode choke not fitted*

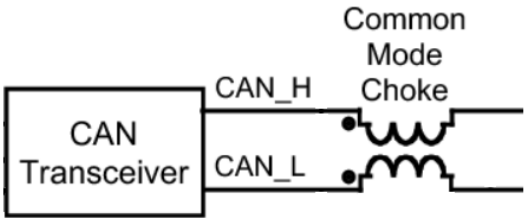



*Figure 11 Common mode choke fitted*

### 3.3.2.9 Capacitors

Between CAN-H and GND and between CAN-L and GND, a capacitor with 47pF/50V shall be used. It shall be ensured that the overall capacitance is not greater than 47pF. The capacitors should be a “match pair” from the same manufacturing batch.
Both capacitors shall be placed close to the bus connector with minimum trace length.

| Parameter | Symbol | Unit | Min | Nom | Max |
|---|---|---|---|---|---|
| Capacitor | $C_H$, $C_L$ | pF | | | 47 |

*Table 11 Capacitors*

#### 3.3.2.10 ESD-Protection

ESD protection is option fit depending on the ESD requirement of the ECU. If zener diodes are required, then the breakdown voltage as stated in the table below.

| Parameter | Symbol | Comment |
|---|---|---|
| Zener diode | $Z_H$/$Z_L$ | 27V breakdown voltage |

*Table 12 Zener diode*

#### 3.3.2.11 External ESD-Protection

External ESD-Protection is not allowed on CAN-Bus-systems.

#### 3.3.2.12 Bus Interface with Wake-Up Function

All ECUs within the CAN network shall implement a CAN-transceiver which is capable to be woken up by a CAN-message. Transceivers which support the wake-up patterns according to ref: [NS1] and [NS2] shall be used.
Deviation is only allowed for carry over parts but that has to be agreed with the EEA team.

#### 3.3.2.13 Controller Oscillator

Only crystals shall be used for clock generation. The tolerance of the crystals shall be less than ± 0.15% over the according temperature and aging ranges. Minimum acceptable clock frequency is 5Mhz.
PLL is not allowed for Vinfast projects.

## 3.4 Data Link Layer

### 3.4.1 Introduction

The data link layer is based on ISO standards and SAE recommendations.

For a more detailed specification on data link layer, this can be found in [NS1], [NS2].

**VCS_001:**
The standard message format with 11-bits identifier shall be used.

**VCS_002:**
Remote frame and overload frame as specified in [NS1] shall not be used.

**VCS_003:**
The DLC in all CAN frames shall be set to 8 bytes.

### 3.4.2 Baud Rate and Bit Timing Parameters

#### 3.4.2.1 CAN transmission speed.

HS-CAN operates at 500kbit/s.
Applicable for all CAN bus: Body_CAN, Info_CAN, Powertrain_CAN, Chassis_CAN.
For given oscillator frequencies, the bit time configuration setup for ECUs connected to a CAN bus at 500kbps are given in table 12.

| Oscillator Clock frequency | Clock Divider | Num of TQs for bit rate | TsyncSeg | Tseg1 | Tseg2 | SJW | Sample position % |
|---|---|---|---|---|---|---|---|
| 5MHz | 1 | 10 | 1 | 7 | 2 | 2 | 80 |
| 8MHz | 1 | 16 | 1 | 12 | 3 | 3 | 81 |
| 10MHz | 1 | 20 | 1 | 15 | 4 | 4 | 80 |
| 12MHz | 2 | 12 | 1 | 9 | 2 | 2 | 83 |
| 16MHz | 2 | 16 | 1 | 12 | 3 | 3 | 81 |
| 20MHz | 2 | 20 | 1 | 15 | 4 | 4 | 80 |
| 40MHz | 4 | 20 | 1 | 15 | 4 | 4 | 80 |
| 80MHz | 8 | 20 | 1 | 15 | 4 | 4 | 80 |

*Table 13 bit time values*

### 3.4.3 Bus-Off

The CAN Driver shall provide functions to handle a bus-off state of the CAN Controller

- In the bus-off state, a node shall not send any frames.

## 3.5 Network Layer

Refer to reference [NS1], [NS2].

### 3.5.1 STmin

The STmin value to be used for the Vinfast project.

| Parameter | OEM enhanced diagnostics | OBD diagnostic | ECU-re Programming (flashing)1 | User or data supplier specific |
|---|---|---|---|---|
| STmin | 20ms | 0ms | 0ms direct on CAN-Bus without Gateway and 5ms through Gateway | 10ms |

*Table 14 STmin values*

## 3.6 Overview of CAN-Network

### 3.6.1 Introduction

This section specifies the Network Management for communication among ECU, used on road vehicles and is to be referenced by every ECU specification that incorporates interconnection for ECU communication. Application layer services are usually referred for example to as diagnostic services, general Network handling or Services for measurement and calibration. In practice, interconnected ECUs from different suppliers influence each other on the same bus, even if they implement CAN-communication according to ISO specifications. A faulty ECU influences other ECUs and the entire system as well. The result is that too many interdependent DTCs in different ECU fault memories aggravate the complexity of system fault analysis. Because this situation has to be solved in the same way for all ECUs, there is a need to standardize network handling as its own functionality and to separate it from ECU- functionality.

Moreover, this so standardized network handling functionality, also called "Network Management", shall fulfil these general requirements:

- Activation/deactivation of the CAN-hardware (controller, bus transceiver) during the startup procedure. Monitoring network configuration while bus system is running.
- Bus diagnosis and providing relevant bus-status information for the whole network.
- Bus error correction/recovery after a bus-off.
- Synchronized transfer of all network nodes to the bus sleep mode.

### 3.6.2 Wakeup/shutdown of ECUs with Ignition on/off

It is some time misunderstood that no network management is required for ECUs, which are activated with "ignition ON". This is true, if all ECUs on the same bus are powered over Ignition line and they can all be powered on at the same time with "ignition ON" and powered off at the same time again with "ignition OFF". It could be easy to survey this situation for two or three ECUs, which are activated/deactivated on the same communication bus with "ignition ON/OFF", but it is no more trivial to handle more ECUs in this way. It is not necessary, that all ECUs shall wakeup at the same time, but a coordinated transition to sleep mode is fundamental and required. Even if ECUs wake up with "ignition ON", they can take different long times on "ignition OFF" to reach their sleep or shutdown modes. Due to their different requirements on and before shutdown, some ECUs need more time to complete their activities like saving data in EEPROM or waiting for completion of slow mechanical and physical processes.

### 3.6.3 Network management and Clamp types.

CL.30 and CL.30s powered ECU shall implement CAN network management.
CL.15 powered ECU shall not implement CAN network management.

### 3.6.4 Network Management Strategies

For Network Management use AUTOSAR-NM of decentralized and direct network management existing in field and on the market, where every NM full node type ECU transmits a special network management message, which is reserved for the network management.

**AUTOSAR-NM**

- Startup / Shutdown of Network
- Controlled transition of all ECU’s to bus-sleep mode
- User data handling

**Network Management General Information**

For the ease of reuse and future development, it is required, that Network management for OEM Vehicles shall be compliant with AUTOSAR-Network Management.

**VCS_004:**

OSEK-NM is not allowed to be used in the ECU.

**VCS_005:**

1. Modules shall use Autosar Classic as the BSW in the module.
2. Autosar v4.3.1 or above to be used as the BSW.

3. Modules can use a mixture of BSW components from various Autosar versions as long as the components are at v4.3.1 or higher. For example, the SecOC from Autosar 4.4 can be used with NM from 4.3.1. EEA team have to be notified of combinations falling this category.
4. EEA team approval required regarding BSW vendor prior to supplier nomination.

### 3.6.5 AUTOSAR-Network Management

The required AUTOSAR Network Management functionality for CAN ECUs in OEM-vehicles is specified in reference [NS5].

**VCS_006:**
Only ECUs that have a reason to demand the network, shall transmit NM frames.
Reference: [NS5] SWS_CanNm_00087.

The reason to be awake shall be agreed with the Networks architecture team.

The following table is used to determine node types and capability to transmit NM frame.

| ECU NM node type | Clamp powered | Description | Transmits NM frame? |
|---|---|---|---|
| Full | CL.30/30s | Can have a valid reason to demand the network to be awake.<br>Can be woken by CAN bus traffic.<br>Typically a function master.<br>Will stay awake depending if receiving NM frames or demanding the network.<br>Will function while awake. | Yes |
| Passive | CL.30/30s | Does not have a reason to demand the network awake.<br>Can be woken by CAN bus traffic.<br>Will stay awake depending if receiving NM frames.<br>Will function while awake. | No |
| None | CL.15 | Does not use CAN NM.<br>Is awake when powered, (transmits CAN frames). | No |

| | | | |
|---|---|---|---|
| | | May use an ignition status message to judge when power off will occur.<br>Will function while awake. | |

*Table 15 NM node types*

**VCS_007:**
Full and Passive ECU shall enter “Bus Sleep Mode” following no NM frames transmitted by any NM ECU and the timers CANNM_TIMEOUT_TIME + CANNM_WAIT_BUS_SLEEP_TIME have expired.
Reference: [NS5] SWS_CanNm_00088.

**VCS_008:**
NM ECUs will have the following operating modes:
**Normal operation state** → All frames transmitted and received as per CAN Database, (usage: NM full node type).

**Ready sleep state** → All frames transmitted and received as per the CAN database, NM frames are NOT transmitted, (usage: NM full node type and NM passive node type).

**Prepare bus sleep mode** → ECU can receive frames only, (usage: NM full node type and NM passive node type). Emptying of buffers is allowed.

**Bus sleep mode** → No frames are transmitted as per the CAN database, (usage: NM full node type and NM passive node type).

The transceiver is in polling mode for network wakeup and/or controller in polling mode for hardware wakeup.

**Diagnostic only mode** (programming session) → Diagnostic frames only, (usage: functional/physical addressed diagnostics for all nodes that support diagnostics).

| **NM mode** | **NM frames transmit/receive ability** | | **Application frames transmit/receive ability** | |
|---|---|---|---|---|
| | *Tx* | *Rx* | *Tx* | *Rx* |
| Bus sleep mode | no | yes | no | yes |
| Prepare bus sleep mode | no | yes | no | yes |
| Normal operation state | yes | yes | yes | yes |

| | | | | |
|---|---|---|---|---|
| Ready sleep state | no | yes | yes | yes |

*Table 16 ECU behaviour for msg Rx & Tx*

**VCS_009:**
Bus load reduction mechanism shall not be used in the Vinfast project.

Partial networking shall not be used in the Vinfast project.

The repeat message state shall not be used in the Vinfast project.

Note: Because the repeat message state is not used, this implies NM burst is not used.

#### 3.6.5.1 AUTOSAR-Network Management States and behaviour

Autosar Network Management is used to provide controlled system wake up, operation and shutdown for the entire system.
Network management is described as a state machine, ref. [NS5].
The figure below and description gives the Vinfast implementation of Network Management.

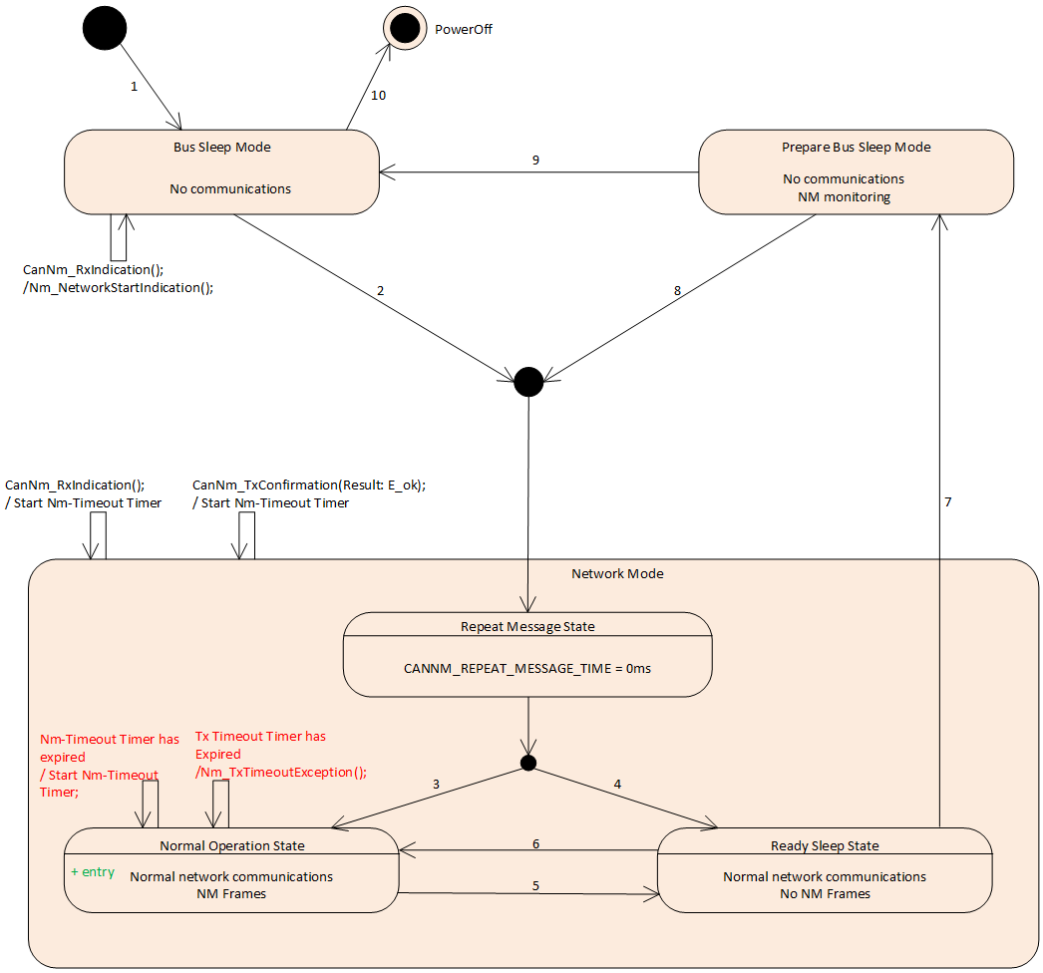



*Figure 12 CAN Network Management Algorithm*

**Transitions:**

**Transition 1:** CL.30s ECU powered on, (or first battery connection CL.30/CL.30s).
**Transition 2:** ECU requires network (internal) or CAN frame received.
**Transition 3:** The network is required, (at least one NM full ECU requires to drive the network awake and transmit NM frames).
**Transition 4:** The network is not required, (NM passive and NM full but does not require to drive the network awake).
**Transition 5:** The network is no longer required, (NM frames were transmitted but are ceased).
**Transition 6:** Network request, (an NM full ECU requires to transmit NM frames).
**Transition 7:** NM Timeout timer expired, (no NM frames are received so timer expires).

**Transition 8:** ECU requires network or NM frame detected.
**Transition 9:** Wait bus sleep timer expired.
**Transition 10:** Power disconnected, (CL.30s or battery disconnect).

**States:**

**Bus Sleep mode:** NM active nodes – network not required/no CAN frames detected. Low power mode. NM passive nodes – no CAN frames detected. Low power mode active.
**Network mode:** initialize/re-initialize CanNmTimeoutTime timer. (NM frames on the bus?)
**Normal Operation State**: NM frames transmitted. At least one NM frame demanding the bus. Functional communications are active.
**Ready Sleep State:** No NM frames are transmitted. NM Timeout timer set. Functional communications are active.
**Repeat Message state:** Not required.
**Prepare Bus Sleep Mode:** Transmit buffers are emptied. No functional communications. NM monitoring active.

#### 3.6.5.1.1 Timing Parameter for AUTOSAR

**VCS_010:**
In the table below the Network management timing parameters, mentioned in the previous sections are listed.

| Parameter | Value |
|---|---|
| CANNM_TIMEOUT_TIME | 2000 ms |
| CANNM_WAIT_BUS_SLEEP_TIME | 1500 ms |
| CANNM_REPEAT_MESSAGE_TIME | 0 ms |
| CANNM_MSG_CYCLE_TIME | 640 ms |

*Table 17 Network management timing parameters*

**VCS_011:**
CANNM_REPEAT_MESSAGE_TIME is set to 0ms, this effectively will bypass Repeat Message State.

CanNmBusLoadReductionEnabled == false sets bus load reduction to off.

#### 3.6.5.1.2 NM configuration Parameter

**VCS_012:**
NM switch configuration values are given in table 17.

| Configuration switch | Value | description |
|---|---|---|
| NM_USER_DATA_ENABLED<br>CANNM_USER_DATA_ENABLED | ON | Enable User Data as required for bus diagnostic |
| NM_BUS_SYNCHRONIZATION_ENABLED<br>CANNM_BUS_SYNCHRONIZA TION_ENABLED | ON | only for ECUs with Gateway Functionality. If an ECU doesn't implement a Gateway-functionality(e.g. NM-Coordinator) this value shall be set to OFF |
| NM_REMOTE_SLEEP_IND_ENABLED | OFF | Except ECUs for which, it is explicitly required to enable "remote sleep indication". |
| NM_NODE_DETECTION_ENABLED,<br>CANNM_NODE_DETECTION_ENABLED | ON | Enable by default to be able to further use the functionality without modification of NM-Configuration |
| NM_CONTROL_BIT_VECTOR_ENABLED<br>CANNM_CONTROL_BIT_V ECTOR_ENABLED | OFF | Disabled. Extension for partial networking. |
| NM_BUS_LOAD_REDUCTION_ENABLED | OFF | Disabled According to NM_BUSLOAD_001.<br>Each ECU send its NM-PDU cyclically if it required network communication |
| CANNM_IMMEDIATE_RESTART | ON | To enable immediate transmission of an NM message if an ECU requires the network in "Prepare Bus Sleep" |

*Table 18 NM configuration switches*

### 3.6.5.1.3 AUTOSAR-Network Management Messages

**VCS_013:**
For network management communication, every node (ECU) shall be assigned a unique identifier (NODE_ID). It is mandatory to use the same identifier for network management and for diagnostics. The node identifier (NODE_ID) shall be an eight-bit value of the range 0x00 to 0x7F.

The node identifier together with the 3-bit class identifier (as defined in section CAN communication for network management) shall be used to determine the CAN identifier for network management message of each NM-Node (ECU).
0x500 < (CAN NM-Message Identifier = 0x500 + NODE_ID) <= 0x57F

**VCS_014:**
The NM frame shall be 8 bytes in length according to the format in figure 18.

| | Bit7 | Bit 6 | Bit 5 | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 |
|---|---|---|---|---|---|---|---|---|
| Byte 7 | User Data 5 (usrData5) | | | | | | | |
| Byte 6 | User Data 4 (usrData4) | | | | | | | |
| Byte 5 | User Data 3 (usrData3) | | | | | | | |
| Byte 4 | User Data 2 (usrData2) | | | | | | | |
| Byte 3 | User Data 1 (usrData1) | | | | | | | |
| Byte 2 | User Data 0 (usrData0) | | | | | | | |
| Byte 1 | Control Bit Vector (CtrlBitVector) | | | | | | | |
| Byte 0 | Source Node Identifier (srcNodeId) | | | | | | | |

*Table 19 Network management message format*

**VCS_015:**
The unique node identifier (NODE_ID) shall be transmitted in byte 0, srcNodeId.

**VCS_016:**
The Control Bit Vector shall be transmitted in byte 1, CtrlBitVector. The control bit vector shall be constructed as given in figure 13.

| | Bit 7 | Bit 6 | Bit 5 | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 |
|---|---|---|---|---|---|---|---|---|
| CBV | reserved | Partial Network Information Bit | reserved | Active Wakeup Bit | NM Coordinator Sleep Ready Bit | Reserved | Reserved | Repeat Message Request |

*Figure 13 Control bit vector in the NM message format*

The Control Bit Vector shall consist of:
**Bit 0**: Repeat Message Request.
0=Repeat Message State not requested.
1=Repeat Message State requested.

**Bit 3**: NM Coordinator Sleep Bit.
0=Start of synchronized shutdown is not requested by main coordinator.
1=Start of synchronized shutdown is requested by main coordinator.
**Bit 4** Active Wakeup Bit.
0=Node has not woken up the network (passive wakeup).
1=Node has woken up the network (active Wakeup).
**Bit 6** Partial Network Information Bit.
0=NM PDU contains no Partial Network request information.
1=NM PDU contains Partial Network request information.

**Bits 1, 2, 5, 7:** These bits are reserved and not used.

**VCS_017:**
User data usage reserved items.
User Data 2, User Data 3, User Data 4 and User Data 5 are reserved and not used.

#### 3.6.5.1.4 Bus Wake-up

**VCS_018:**
Network nodes woken-up via external events that require bus communication shall transmit a network management message to signal its intention to communicate on the Bus. A CAN transceiver with wakeup functionality that observes the wake-up pattern on the bus shall activate the ECU - application, if all activation conditions for ECU are fulfilled. The ECU-application shall transmit a network management message in this case.

**VCS_019:**
Due to differing implementations of CAN transceivers, a CAN transceiver may also activate the ECU on shorter dominant states observed on the bus. To prevent ECUs from unintentional wake-up, the dominant time for wake-up via bus of the used transceiver is defined to be between 0.75µs and 5µs.

**VCS_020:**
In NM-PDU, there are 6 bytes called "User Data" reserved for user or OEM specific usage. They are supplementary ECU-application specific pieces of data that are attached to every NM message sent

on the bus. The usage of the user data is not standardized, so their usage can be a limitation for future extensions or modifications. User data is helpful for analysis of global fault of ECU startup or sleeping instances and therefore, they shall be documented in the communication matrix.

**VCS_021:**
User data support shall be enabled for each NM full node type.

**VCS_022:**
User data shall be read and written at application level using the functions specified in SWS_CanNm_00035, SWS_CanNm_00132 in [NS4].

- Nm_GetUserData(…)
- Nm_SetUserData(…)

**VCS_023:**
By default, byte 4-7 of user data (table 18) shall be set to 0x00 indicating, that they are not used for a specific purpose.

**VCS_024:**
The first user data byte **usrData0** indicates the implemented functionality of user data. This value is to be acted as a version number, that tell how to interpret the content of the remaining five user data bytes (usrData1, usrData2, …, usrData5).
This value is always constant for given software version.

| usrData0 | Description/Interpretation |
|---|---|
| 0xFF | User data not used [not to be used] |
| 0x00 | User data for current vehicle without partial networking |
| 0x01 | User data for current vehicle with partial networking [not to be used] |

*Table 20 Interpretation of usrData0 as functionality of user data*

**VCS_025:**

The “wakeup reason” and “reason to be awake” shall be provided in **usrData1** from the application of every ECU supporting network management at any time when the network is awake.

The ECU transmits the application reason that caused the CAN network to wakeup. The bit needs to be cleared when the NM transitions to the Normal mode. The application then needs to log the reason for keeping awake as per table 21.

For example, bit 1 and bit 5 cannot be set simultaneously. The wakeup reason can be ignition apply but that bit needs to be reset post wakeup and bit 5 needs to be set to indicate that ignition is the reason for keeping awake. The changeover is to be done in the Normal Operation State

| usrData1 | Definition | Description/Interpretation |
|---|---|---|
| Bit 0 | IGNITION_WAKEUP<br>0x0 inactive<br>0x1 active | ECU was "power off" or "Bus sleep" mode and started by Ignition turn on. |
| Bit 1 | RESET_WAKEUP<br>0x0 inactive<br>0x1 active | If ECU was already awake, in "Prepare Bus Sleep" or "Ready Sleep" mode and re-started by an internal or an external reset. |
| Bit 2 | NETWORK_WAKEUP<br>0x0 inactive<br>0x1 active | ECU was in "Power off", "Bus Sleep", "Prepare Bus Sleep" or "Ready Sleep" Mode and then re-started by a network communication. |
| Bit 3 | ECUSPEC_WAKEUP<br>0x0 inactive<br>0x1 active | ECU was in "power off", "Bus Sleep", "Prepare Bus Sleep" or "Ready Sleep" mode or "awake" and re- started by ECU-specific reason (e.g. pressed Button, timer, etc.). |
| Bit 4 | NETWORK_AWAKE<br>0x0 inactive<br>0x1 active | ECU need network communication and keeps the network awake. |
| Bit 5 | IGNITION_AWAKE<br>0x0 inactive<br>0x1 active | ECU can’t release communication because Ignition is turn on. |
| Bit 6 | DIAGNOSTIC_AWAKE<br>0x0 inactive<br>0x1 active | ECU need communication for diagnostic purpose and keeps network awake. |
| Bit 7 | ECUSPEC_AWAKE<br>0x0 inactive<br>0x1 active | ECU keeps the network awake for ECU-define reasons for which communication is needed. |

*Table 21 Defined wake up reasons*

The ECU shall clear any set bits when it transitions out from Normal Operation State.

### 3.6.6 NM at Gateways

Each vehicle has a central gateway as designated by the EEA team. This module will be designated to hold the NM-Coordinator component as defined in the Autosar NM specification [NS5]. The central gateway is the module connected to all communication buses designated by the EEA team as individual domain backbones. The NM implementation would be multi-instance with a piece of code monitoring the following

1. The NM coordinator has to monitor the NM frames on all channels and keep the vehicle awake as long as there is a single module that is sending NM frames indicating that it needs the vehicle network. The implementation will be in normal state and sending NM frames on all channels to keep vehicle awake.

2. When there is no module on the vehicle which needs the vehicle network then the coordinator will transition from Normal state to Prepare bus sleep state and will stop sending NM frames on each channel. This is in preparation to eventual transition to Bus sleep state.

### 3.6.7 NM and Diagnostic communication control services

As required in NM_GENERAL_001, ECU shall transmit periodic NM PDUs (Messages) as long as it requires bus-communication, but in some situation, it is required for programming purpose for example, to disable normal communication including NM-Messages transmission during a diagnostic programming session. After Programming session, it is necessary to enable again the normal communication and NMmessages transmission.

**VCS_026:**

To disable and re-enable NM-messages in diagnostic sessions, the AUTOSAR-NM provides two API-Functions that shall be used for this purpose. See SWS_CanNm_00033 in [NS4].

- Nm_DisableCommunication(…)
- Nm_EnableCommunication(…)

## 3.7 CAN Wakeup/Sleep

### 3.7.1 Introduction

To ensure the operational readiness of the overall system (whole vehicle) in a reasonable time, it is required for each ECU in the vehicle communication network to have a maximum start-up time as specified.

Conversely the vehicle shall also have shut down timings for the communication network.

#### 3.7.1.1 Wake up

Wakeup timing can be catagorised by the following two figures.

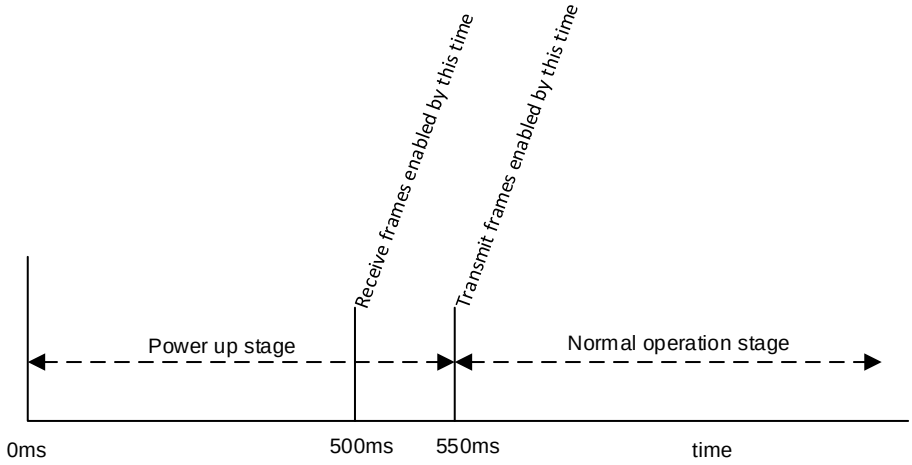



*Figure 14 Power on wakeup*

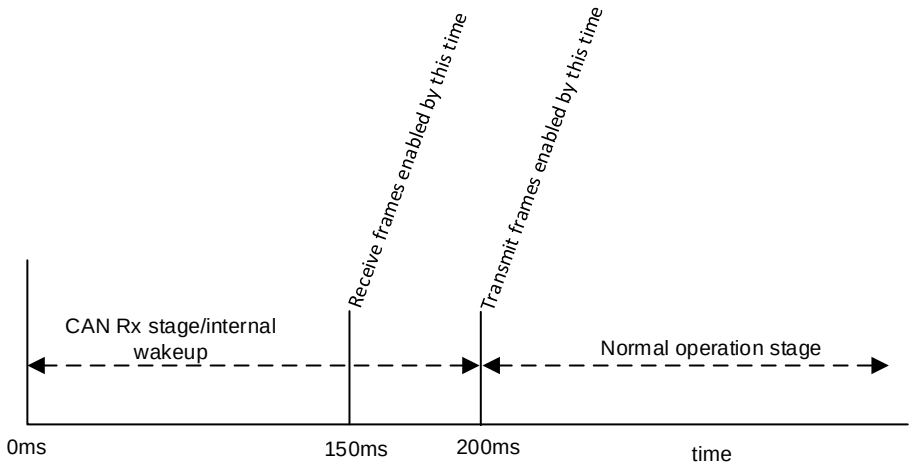



*Figure 15 Wakeup by CAN frame received or internal wakeup*

### 3.7.1.2 Bus sleep

The figure below shows the idealised minimum time to achieve Bus sleep mode.
In a real system an NM ECU may have local application timers could that influence the entering into Bus sleep mode.

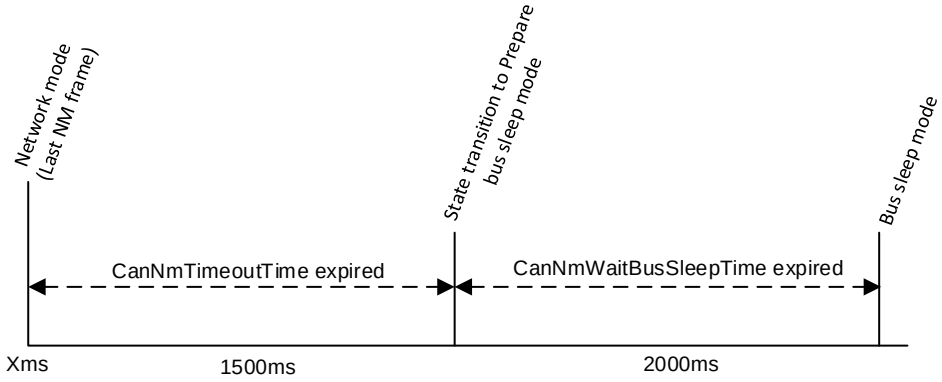



*Figure 16 Entering sleep mode*

### 3.7.2 Safety and availability requirements

**VCS_27:**
Every network node (ECU) shall ensure/guarantee proper communication for power supply voltage range defined in [QD10]. section Operation voltage ranges.

**VCS_28:**
A high bus load on the bus shall not disturb the ECU functionality or put the ECU in an unsafe state. Please note that peak theoretical load on the CAN bus will not exceed 70%. Theoretical load is calculated at design state as a static model with the assumption that all modules are trying to transmit at the same time during the e-motor running state (as defined in the clamp specification, [QD12]).

### 3.7.3 Acceptance of RX-messages

Each ECU shall only receive and monitor its received, (RX) - signals as specified in the communication matrix (dbc file). This means, that no message shall pass the PDU router, if its content is not needed in the ECU-application. Each transmitted frame needs to be assigned a dedicated hardware slot (memory buffer) and the remaining hardware slots are to be assigned to the highest priority frames. If there are more received frames than available slots, then a specific slot can be shared between the received frames and an interrupt mechanism to be used to clear the slot on reception of the designated frame serviced by interrupt.

## 3.8 CAN-Communication

Bus inputs are defined as signals in messages received by an ECU from the bus and shall be interpreted as specified in the communication matrix. The following requirements apply to ECU bussed inputs.

### 3.8.1 Signal Specification

A signal is a bus input and indicates a single part of CAN-message, defined by a resolution in bits and a formula to calculate the value. For representation of signal values, different data types as specified bellow are used.

#### 3.8.1.1 Bit Order

According to the CAN bus protocol, the bit order is a **big endian mode**, that is, **msb first**, so there is no need to distinguish. Motorola format is to be used for the signal set to be defined in the CAN database (*.dbc file).

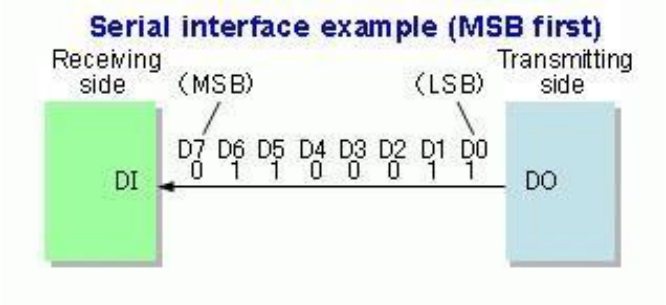



*Figure 17 Example for most significant bit*

#### 3.8.1.2 Byte Order

**VCS_29:**

All signals shall be represented in **Motorola forward MSB** format (big endian). That means the most significant byte (MSB) shall be transmitted at first and the least significant byte (LSB) shall be transmitted at last.

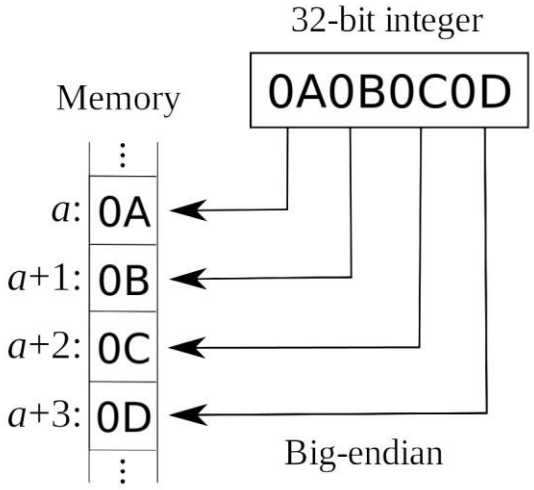



*Figure 18 Example of Motorola byte order*

#### 3.8.1.3 Data Types

On CAN different data types with varying size may be used.

##### 3.8.1.3.1 Unsigned Integer

**VCS_30:**

Unsigned integers are to be used. Signed integers and floating point are not allowed.

#### 3.8.1.4 ASCII-Code

**VCS_31:**

Signal types need to be enumerated as value range or physical range. Bit coding is not allowed.

1. Value range is where each value of the signal has a state description depicting a condition.
2. Physical range is where a range of values indicate a calculated value from a function.
3. Combination of the above is also allowed.

Signals with no encoding are only allowed for the following.

1. Diagnostics frames.
2. Network management frame.
3. Development frames.

**VCS_32:**

Signals should have the following characteristics.

1. Initialisation value
2. The complete value range should have a description. No undefined range is allowed.
3. SNA value refer to section 3.8.1.5.

**VCS_33:**

The "Signal Initial Value = SIV"-value shall be used at network startup by a transmitting ECU when a signal is transmitted before a valid value has been measured or computed.

#### 3.8.1.5 Signal not available

The vehicle communication matrix shall define a "Signal Not Available" -SNA- value. For example for some signals computed or routed from Gateway. If an SNA is defined for a given signal, the

application software should set its transmitted value to SNA when the input is un-known or becomes unavailable. Some situations where SNA may be used are when a sensor fails or a subsidiary bus, such as LIN, fails leaving the actual value of the signal unknown to the device that transmits the information on the CAN communication bus.

**VCS_34:**
Unless otherwise defined, "Signal Not Available = SNA" value is defined as following: SNA-Value = SNV–Value.

#### 3.8.1.6 Empty or unused bits.

Empty or unused bits shall be padded with ones (1).

#### 3.8.1.7 Gateway specific usage of SNA

This is applicable only for signal routing.

For signals that are to be routed via “signal routing” function in a gateway, a SNA state has to be defined in the encoding. The gateway application will absorb the signal and will monitor the signal for the timeout period.

If the gateway does not receive a fresh instance of the incoming frame which contains the signal for the monitoring period, then the gateway will set the value of the signal in the output frame as the SNA state.

That state is to be used as indicator by the receiver that the original transmitting module on the other side of the gateway has stopped sending data on the CAN bus.

If SNA is not available as where off the shelf modules are being integrated into the vehicle, so the gateway will set the value of the signal to an alternative like the default value or indicative error value in the encoding. This will have to be agreed by all parties prior to implementation.

#### 3.8.1.8 Message Classes

A Message is the effective part of a Frame, that can be used by the application. They are raw bytes with max length and identifier as defined in the protocol specification. In addition to general specification, they are OEM specific message properties specified in the communication like message classes, categories, etc.
The CAN-messages of the following type are allowed on the network

1. Cyclic frames.
2. Cyclic frames with repeat rate.
3. Event triggered frames.

### 3.8.1.9 CAN Identifier Scheme

There are different messages types for different types of data (effective application, network management, diagnostic, transport protocol using by Multimedia and/or Telematics, etc.)

| Range-ID | CAN- Messages Identifiers range | | Message group Priority | Description |
|---|---|---|---|---|
| | Start/Base | End | | |
| CANID_R00 | 0x001 | 0x4FF | - | To be used for normal data applications |
| CANID_R01 | 0x501 | 0x57F | - | Network management messages Autosar |
| CANID_R02 | 0x580 | 0x5FF | - | Unsegmented messages transmission for special functions |
| CANID_R03 | 0x601 | 0x6FF | - | UDS diagnostics for the modules |
| CANID_R04 | 0x700 | 0x7DE | - | Module development frames plus calibration |
| CANID_R05 | 0x7DF | 0x7EF | - | Legislated OBD/EOBD diagnostics |
| CANID_R06 | 0x7F1 | 0x7F1 | - | Airbag scrapping ISO 26021 |
| CANID_R07 | 0x7F2 | 0x7F8 | - | Reserved |
| CANID_R08 | 0x7F9 | 0x7F9 | - | Airbag scrapping ISO 26021 |
| CANID_R09 | 0x7FA | 0x7FF | - | Reserved |

*Table 22 CAN Identifier Classes*

### 3.8.1.10 Message Transmit Types

**VCS_35:**

Message transmit type describes when, or under which, conditions a message shall be transmitted. A network node is allowed to send a message at any time, but in general only two causes for a message transmission occur: A message may either be sent periodically using a fixed time interval, or the transmission may be triggered by some event.

#### 3.8.1.10.1 Cyclic Messages

A "Cyclic" message shall be transmitted on a regular periodic schedule or fixed time, independently of changes in the signals contained within the message. The exact time between two transmissions, meaning the jitter of a cyclic message, may vary depending on the bus load.

| Value | Minimum | Nominal | Maximum |
|---|---|---|---|
| $t_{cycle}$ | 10ms | - | - |
| Timeout factor | 2 | 5 | 10 |

*Table 23 Cyclic Message Properties*

A node that receives a cyclic message shall observe the cycle time. If a cyclic message is not received within a certain amount of time defined as timeout, the receiving node shall take appropriate actions according to its specification document. The timeout for a cyclic message shall be defined globally in the vehicle communication matrix (e.g., 5 times the cycle time of the message).

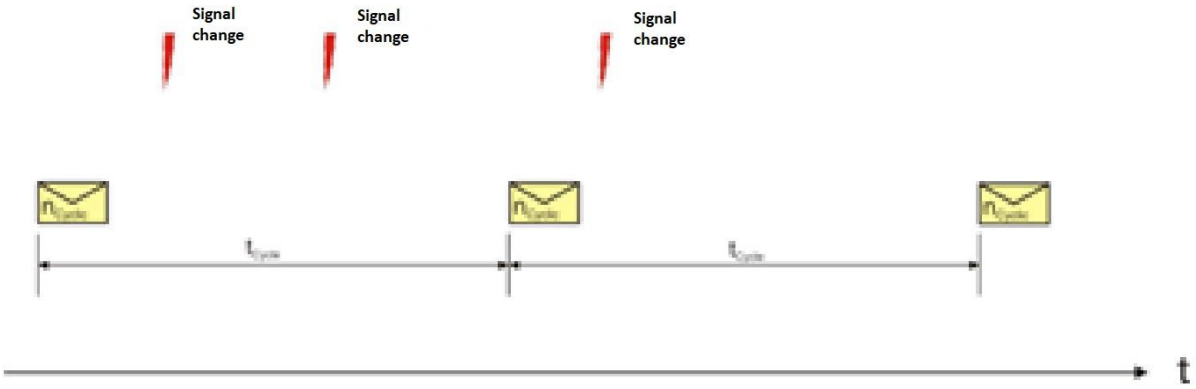



*Figure 19 Timing diagram cyclic message*

The average cycle time of every cyclic message shall be within ± 2% of its defined cycle time.

#### 3.8.1.10.2 Request Messages

Request messages are transmitted after request from a service message of another ECU. If the answer is not received within $t_{wait}$, the ECU will request the message again.

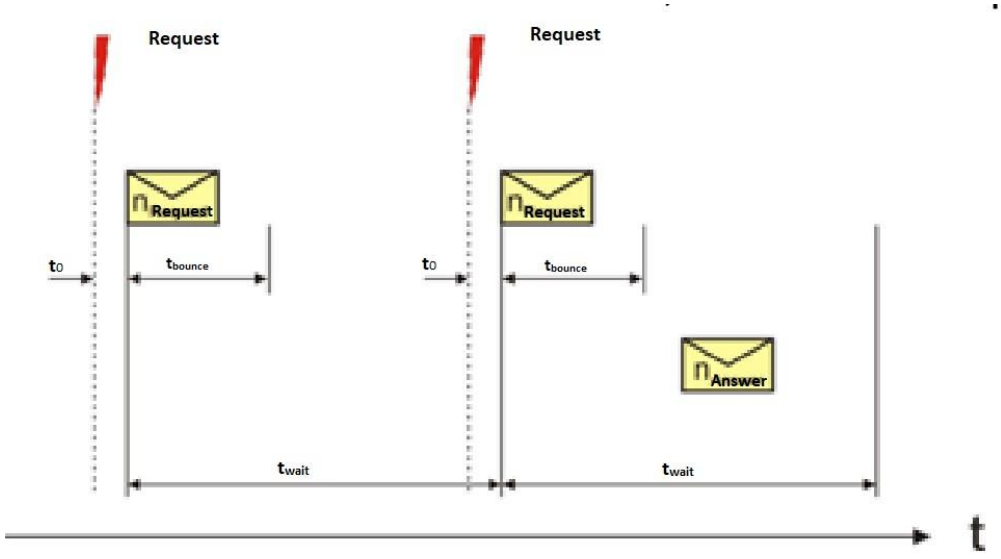



*Figure 20 Timing diagram request message*

### 3.8.1.10.3 Event Triggered Messages

Event triggered messages are only transmitted on the bus if the according event occurs.

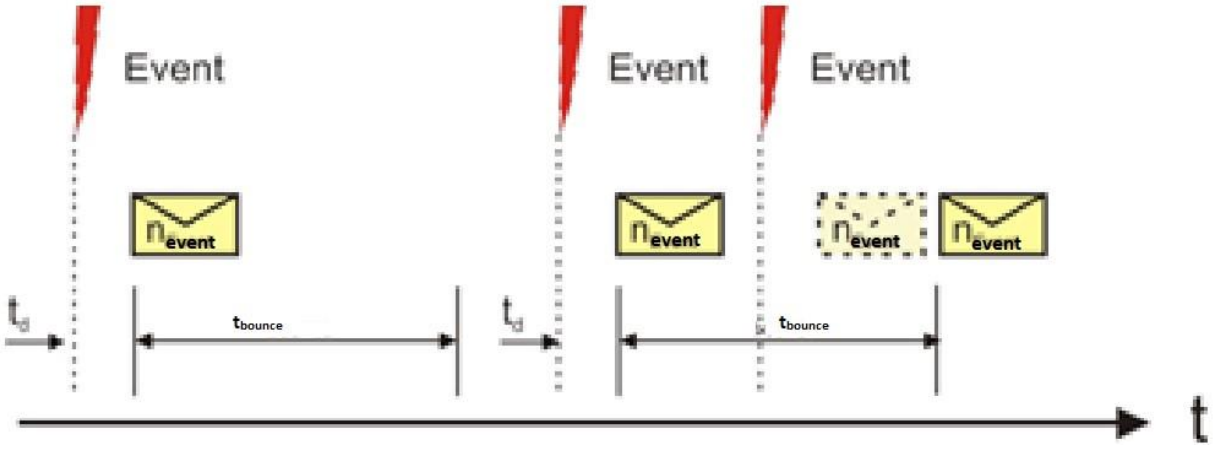



*Figure 21 Timing diagram event triggered message*

If a second event occurs within $t_{bounce}$ of the first event the according message shall be sent after $t_{bounce}$ of the first message has expired. The minimal time $t_{bounce}$ between two consecutive transmissions of an event triggered message shall be defined globally in the vehicle communication matrix. The tolerance of $t_{bounce}$ is given in table 23.

| Value | Minimum | Nominal | Maximum |
|---|---|---|---|
| $t_{bounce}$ | 4ms | 5ms | 10ms |

*Table 24 Tolerances of event interval time tmin*

#### 3.8.1.10.4 Event Triggered with Repetition

In that special case after the message, triggered by an event, is sent, the message shall be sent another e.g. 2-times with a period of e.g. $t_{Rep}$=100ms due to security reasons (Details specified in CAN-DBCs).

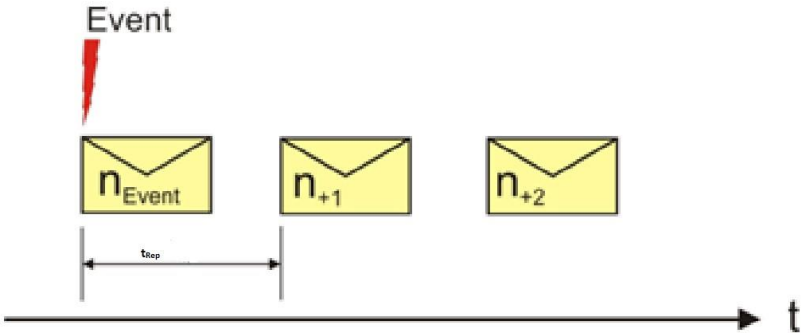



*Figure 22 Timing diagram event triggered message with repetition*

#### 3.8.1.10.5 Cyclic with Repetition on Event

In that special case after the message, triggered by an event, is sent, the message shall be sent another e.g. 2-times with a period of $t_{Rep}$=100ms due to security reasons (Details specified in CAN-DBCs). Additionally, to that the message shall be sent cyclic with cycle time $t_{Cyclic}$ which is usually high (e.g. 1000ms).

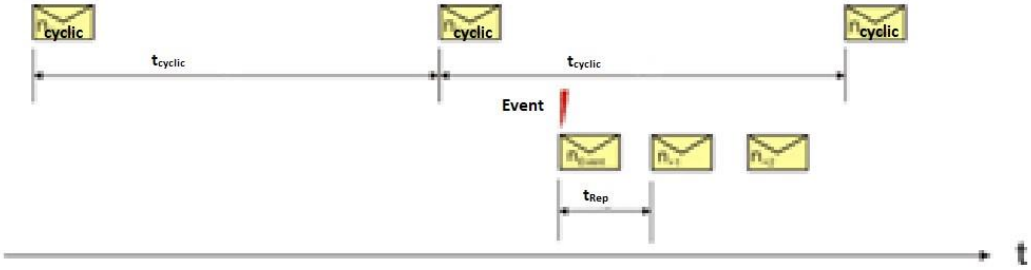



*Figure 23 Timing Diagram Event Triggered with Repetition and Cyclic Message*

#### 3.8.1.11 Protection of Safety Messages Across Network

Safety-related data exchange shall be protected at runtime against the effects of faults within the communication link (e.g. from ECU-X on Bus-A trough Gateway to ECU-Y on Bus-B). Examples for such faults are random HW faults (e.g. corrupt registers of a CAN Controller), interference (e.g. due to EMC), and systematic faults within the software implementing the communication stack (CAN-driver

and other communication software components). In order to ensure freedom from interference by software partitioning for example, measures to detect the following communication faults are taken into account, see [NS4] and [NS9].

- loss of peer to peer communication.
- unintended message repetition due to the same message being unintentionally sent again.
- message loss during transmission.
- insertion of messages due to receiver unintentionally receiving an additional message, which is interpreted to have correct source and destination addresses.
- sequencing due to the order of the data being changed during transmission, i.e. the data is not received in the same order as in which it was been sent.
- message corruption due to one or more data bits in the message being changed during transmission.
- age delay due to the message being received correctly, but not in time.
- blocking access to data bus due to a faulty node not adhering to the expected patterns of use and making excessive demands for service, thereby reducing its availability to other nodes, e.g. while wrongly waiting for non-existing data.

Depending on the security requirements of the message which should be protected, there are two methods, which can be combined in any order (checksum and/or alive counter).

**VCS_36:**
For development stage ECUs which implement one or both mechanisms for protection of message content, shall provide a possibility (e.g. on diagnostic service request) to disable and re- enable checking of CRC and/or Alive counter. This is necessary to avoid many CRC- or Alive counter failures in the early FRSs. This feature shall be recorded in the ODX file and shall be switched off for production release software.

#### 3.8.1.11.1 Alive Counter

**VCS_37:**
The actuality of the information which should be protected shall be ensured by the sender. Every sent message should increase the alive counter according to specification (0-14).
Mechanisms that can be used are as follows.

If the value of the alive counter of a message did not change since the last transmission for a configurable number (e.g., 2-5 times, have to be defined separately at each receiver), the message shall be considered as invalid (Default values have to be defined at each receiver). If the value of the counter received has incremented and the delta is within an acceptable range of value. For example, if the value received in first instance of the frame is 1, then the value of the second instance needs to be no more than 2 increments of the counter. So it should not be more than 3.

#### 3.11.1.9.1 Application CRC

**VCS_38:**

The application CRC shall be calculated according to the specification of [NS10], CRC- 8 algorithm, which is typically quite efficient and may use hardware support. CRC-8 in [NS10], requires that the initial value and XOR value for CRC calculation is 0xFF.

#### 3.8.1.11.2 Sample C Code of CRC-8 implementation according to SAE J1850

```
#define CRC8_TABLELENGTH 256u
#define UINT8_MAX ((uint8) 0xFF ) /*
CRC8 according to SAE J1850 */
static const uint8 CRC8_J1850_TABLE[CRC8_TABLELENGTH] =
{
0x00 , 0x1D , 0x3A , 0x27 , 0x74 , 0x69 , 0x4E , 0x53 ,
0xE8 , 0xF5 , 0xD2 , 0xCF , 0x9C , 0x81 , 0xA6 , 0xBB ,
0xCD , 0xD0 , 0xF7 , 0xEA , 0xB9 , 0xA4 , 0x83 , 0x9E , 0x25
, 0x38 , 0x1F , 0x02 , 0x51 , 0x4C , 0x6B , 0x76 ,
0x87 , 0x9A , 0xBD , 0xA0 , 0xF3 , 0xEE , 0xC9 , 0xD4 ,
0x6F , 0x72 , 0x55 , 0x48 , 0x1B , 0x06 , 0x21 , 0x3C ,
0x4A , 0x57 , 0x70 , 0x6D , 0x3E , 0x23 , 0x04 , 0x19 ,
0xA2 , 0xBF , 0x98 , 0x85 , 0xD6 , 0xCB , 0xEC , 0xF1 ,
0x13 , 0x0E , 0x29 , 0x34 , 0x67 , 0x7A , 0x5D , 0x40 ,
0xFB , 0xE6 , 0xC1 , 0xDC , 0x8F , 0x92 , 0xB5 , 0xA8 ,
0xDE , 0xC3 , 0xE4 , 0xF9 , 0xAA , 0xB7 , 0x90 , 0x8D ,
0x36 , 0x2B , 0x0C , 0x11 , 0x42 , 0x5F , 0x78 , 0x65 ,
0x94 , 0x89 , 0xAE , 0xB3 , 0xE0 , 0xFD , 0xDA , 0xC7 ,
0x7C , 0x61 , 0x46 , 0x5B , 0x08 , 0x15 , 0x32 , 0x2F ,
0x59 , 0x44 , 0x63 , 0x7E , 0x2D , 0x30 , 0x17 , 0x0A ,
0xB1 , 0xAC , 0x8B , 0x96 , 0xC5 , 0xD8 , 0xFF , 0xE2 ,
0x26 , 0x3B , 0x1C , 0x01 , 0x52 , 0x4F , 0x68 , 0x75 ,
0xCE , 0xD3 , 0xF4 , 0xE9 , 0xBA , 0xA7 , 0x80 , 0x9D ,
0xEB , 0xF6 , 0xD1 , 0xCC , 0x9F , 0x82 , 0xA5 , 0xB8 ,
0x03 , 0x1E , 0x39 , 0x24 , 0x77 , 0x6A , 0x4D , 0x50 ,
0xA1 , 0xBC , 0x9B , 0x86 , 0xD5 , 0xC8 , 0xEF , 0xF2 ,
0x49 , 0x54 , 0x73 , 0x6E , 0x3D , 0x20 , 0x07 , 0x1A ,
0x6C , 0x71 , 0x56 , 0x4B , 0x18 , 0x05 , 0x22 , 0x3F ,
0x84 , 0x99 , 0xBE , 0xA3 , 0xF0 , 0xED , 0xCA , 0xD7 , 0x35
, 0x28 , 0x0F , 0x12 , 0x41 , 0x5C , 0x7B , 0x66 ,
0xDD , 0xC0 , 0xE7 , 0xFA , 0xA9 , 0xB4 , 0x93 , 0x8E ,
0xF8 , 0xE5 , 0xC2 , 0xDF , 0x8C , 0x91 , 0xB6 , 0xAB , 0x10
, 0x0D , 0x2A , 0x37 , 0x64 , 0x79 , 0x5E , 0x43 ,
0xB2 , 0xAF , 0x88 , 0x95 , 0xC6 , 0xDB , 0xFC , 0xE1 ,
```

```
0x5A , 0x47 , 0x60 , 0x7D , 0x2E , 0x33 , 0x14 , 0x09 ,
0x7F , 0x62 , 0x45 , 0x58 , 0x0B , 0x16 , 0x31 , 0x2C ,
0x97 , 0x8A , 0xAD , 0xB0 , 0xE3 , 0xFE , 0xD9 , 0xC4
} ;
/*
* FUNCTION: uint8 crclib_getCRC8(const uint8 *xData, uint16 xDataLen)
* *
PARAMETERS: xData: Pointer(address) of data that CRC8 to be calculated
* xDataLen: size of data (number of bytes)
*
* RETURN: 8 Bits CRC- value
* *
DESCRIPTION: this function calculates the CRC8 value of given data
* according to SAE J1850
*/
uint8 crclib_getCRC8( const uint8 * xData, uint16 xDataLen)
{
uint8 idx;
uint8 crc = UINT8_MAX;
/* calculate CRC8 conform to SAE J1850 */
for (idx = 0 ; idx < xDataLen; idx ++ ) { crc =
CRC8_J1850_TABLE[crc ^ (xData[idx])];
}
/* the calculate CRC value shall be XOR with TUINT8_MAX ! */
crc ^= UINT8_MAX; return
crc;
} /* End of crclib_getCRC8(...) */
```

#### 3.8.1.11.3 Message Layout with End-to-End protection

**VCS_39:**

According to requirements COM_E2EALIVE_001 and COM_E2ECRC_001, each message, which content is to be protected with alive counter or/and CRC, shall define the two signals ALIVE_XXX and CRC_XXX. The suffix XXX in the signal name is equal to the name of Message. The byte 0 of the message shall be reserved for CRC and the less significant nibble (bit 0 … bit 3) of byte 1 shall be reserved for the alive counter.

| Byte | 7 | 6 | 5 | 4 | 3 | 2 | 1 | | | | | | | | 0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit | … | | | | | | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Signal | other Signals… | | | | | | **ALIVE**_XXX | | | | | | | | **CRC**_XXX | | | | | | | |

*Table 25 Layout of Message with Content Protection*

**VCS_40:**
Even if both signals ALIVE_XXX="alive counter" and CRC_XXX="CRC8 value" are always defined in a message with protected content, both protection mechanisms can be used independently from each other. This means, that the content of a message can be protected only with CRC without computation or checking of the alive counter. Protecting a message content with alive counter does not obligate the use of computation or checking of CRC.

**VCS_41:**
The size of the signal ALIVE_XXX is 4 bits, hence it can take values between 0x0 and 0xF where the value 0x0F=15, equal to SNV, indicating that the alive counter is invalid.

**VCS_42:**
The size of signal CRC_XXX is 8 bits. There is no invalid value for invalidation of a CRC-signal embedded in a message with protected content as specified in VCS_38. The whole content of the protected message excluding the CRC (byte 0) itself and beginning with byte 1 to byte DLC-1 shall be used in the CRC-calculation.

## 3.9 Communication Error Handling

Electrical circuit continuity malfunctions (e.g. circuit shorted low, circuit shorted high, and open circuit) are typically caused by a wiring or connector problems, whereby electrical circuit performance/rationality malfunctions are typically more representative of an actual component failure. Network diagnostic is defined as a system level approach to address common failures across multiple ECUs for monitoring and verifying electrical circuit continuity and functional performance of network communication systems. Most automotive electrical/electronic system faults can be divided into two major categories, network and power distribution. Network systems can be classified into four major failure groups representing the integrity of the bus electrical system and its associated data. Power distribution failures can be classified into a single failure group indicating the status of the power supply to an ECU and its associated components. Power distribution faults are mentioned in this specification document because they also can cause network failures in most cases. Failure Detection is defined as the ECU’s ability to identify failure conditions that affect the performance of the ECU or any associated circuits, components, or systems.

### 3.9.1 Network Electrical Failures

Network Electrical failures identify an ECU internal network electrical error, a network circuit short fault condition or a network circuit open fault condition.

**VCS_43:**
Since the CAN network uses a 2-wire twisted pair implementation, a specific DTC shall be defined for each possible circuit short and circuit open condition.

| **Network Failure mode** | **Description** | **Implementation** |
|---|---|---|
| General Bus Failure | Defines a general communication network hardware malfunction (may indicate any electrical circuit continuity failure condition). A DTC for this Failure case (“General Bus Failure”) shall be stored if following conditions are fulfilled:<br>·“no Busoff-indication”,<br>· “no short circuit to ground or to Battery,<br>· “no open circuit (CAN_high, Can_low )”,<br>· “no TX-messages sent for a defined time(e.g. 1000ms)”<br>· “no RX-messages received for a defined time(e.g. 1000ms)” | mandatory |
| CAN High connected to CL31 | Defines a signal value measured at the CAN High communication line that is at or near zero scale (short to ground condition). | optional - mandatory if transceiver is able to detect |
| CAN Low connected to CL31 | Defines a signal value measured at the CAN Low communication line that is at or near zero scale (short to ground). | optional - mandatory if transceiver is able to detect |
| CAN High connected to CL30 | Defines a signal value measured at the CAN High communication line that is at or near full scale (short to battery condition). | optional - mandatory if transceiver is able to detect |
| CAN Low connected to CL30 | Defines a signal value measured at the CAN Low communication line that is at or near full scale (short to battery condition). | optional - mandatory if transceiver is able to detect |
| CAN High Circuit Open | Defines a signal value measured at the CAN High communication line that is inactive (no response or interrupted condition). | optional - mandatory if transceiver is able to detect |
| CAN Low Circuit Open | Defines a signal value measured at the CAN Low communication line that is inactive (no response or interrupted condition). | optional - mandatory if transceiver is able to detect |
| CAN High and CAN Low Shorted | Defines a signal value measured at both the CAN High and CAN Low communication lines that is at the same voltage potential (shorted together condition). | optional - mandatory if transceiver is able to detect |

| | | |
|---|---|---|
| Single-Wire Mode | Defines the reception of messages received from one or more ECU(s) on only a single wire, even though both wires seem to be functioning correctly. Single w ire operation is in general possible for low speed CAN-busses (up to 125 Kbps) and only if the used CAN-transceiver is fault tolerant. | optional - mandatory if transceiver is able to detect |

*Table 26 Network electrical failures*

#### 3.9.1.1 Network Bus-Off Failure

Condition where all bus message transmission has been disrupted and the ECU is logically disabled from the communication network to prevent further bus behavioural disruptions and potential permanent damage to the network.

**VCS_44:**

Every ECU shall be able to detect and report Bus-Off failure as described in [NS1], [NS2], [NS3] and section CAN_DataLink_Layer.

| Network Failure mode | Description |
|---|---|
| Bus off | Defines the status information about failure conditions of the transceiver (e.g. Bus-Off) provided by the communication controller.<br>This includes all failure modes detectable and reported by the bus controller which are not covered by any of the other failure modes and not related to the electrical/physical integrity of the bus. |

*Table 27 Network failure mode bus off*

#### 3.9.1.2 Network Node Integration Failures

Network node integration failures identify whether or not an ECU that is designed for the vehicle is actually communicating on the network. Network node integration fault - detection improves both vehicle build confidence and network fault isolation. A Network node integration failure may indicate a missbuild condition, a configuration error, an electrical hardware failure condition, or an ECU internal failure.

**VCS_45:**

Every ECU shall be able to detect and report following node integration failures in the communication network with exception of "Additional ECU Detected".

#### 3.9.1.3 Message and Signal Content Failures

Message and signal content failures are used to flag data integrity issues associated with a received CAN message or signal. The message and signal content failure may be an intentional signal transmitted by the originating ECU or gateway ECUs rather than a network faulT condition.

**VCS_46:**
Every ECU shall be able to detect and report following message and signal failures:

| Network Failure mode | Description |
|---|---|
| Implausible Signal Received from ECU 'X' | Defines a signal element received through a bus message with 'implausible or invalid data' indicating signal available but with loss of integrity (permanent condition due to component failure condition). |
| Implausible message content Received from ECU 'X' | Defines a message content with 'implausible or invalid data' indicating signals available but with loss of integrity (permanent condition due to CRC- or alive counter - error detected). |

*Table 28 Network failure mode message and signal content failures*

### 3.9.1.4 System Power Supply and Communication Error Detection

Power Distribution failures are used to flag when a supply voltage range exceeds specification.

| Network Failure mode | Description |
|---|---|
| System Voltage too Low | Defines a signal value received as a bussed input which falls below the lower limit of the system voltage range. Supply voltage <9v |
| System Voltage High | Defines a signal value received as a bussed input which exceeds the upper limit of the system voltage range. Supply voltage >16v |

*Table 29 Network failure mode system power supply failures*

**VCS_47:**
On detection of these two failures, "System Voltage High" or "System Voltage Low", no communication failure as defined in section "Network Bus-Off Failure" and "Message and signal content failures" shall be reported or retained, even if they are detected during system voltage failure.

### 3.9.1.5 Ignition State and Communication Error Detection

Network Communication failure detection is performed only with the ignition in the "ON" state. The ignition state CL15 is used as enable condition for Network Communication failure detection. As not all

ECUs are battery fed, these ECUs for example connected on HS-CAN will not transmit messages when the ignition state does not equal "ON".

Note: The bussed input requirements, as specified in section CAN- Communication, are used with the following enable condition requirements.

**VCS_48:**
The ignition state CL15 received over the bus shall take precedence over direct ignition state measurement to achieve standardized ignition state fault handling for all ECUs in the OEM-vehicle.

**VCS_49:**
If the bus communication has failed or an invalid ignition state signal is received, and a locally measured ignition state is available, the locally measured ignition state shall be used.

**VCS_50:**
If the bus communication has failed or an invalid ignition state signal is received, and a locally measured ignition state is not available, the last known valid Ignition state (CL15) shall be used for a configurable amount of time and then shall be considered as OFF in the current power cycle, if the above denoted situation persists.

**VCS_51:**
If the bus communication has failed and a valid ignition state signal has not been received within the current power cycle, and a locally measured ignition state value is not available or valid, the ECU shall assume that the Ignition state is OFF and therefore the enable condition is not fulfilled for Network Communication failure detection.

# 4. Development Environments

Not relevant.

# 5. References

## 5.1 Quoted Documents

| Ref. No. | Name | Version |
|---|---|---|
| QD1 | Basic_Diagnostic_Specification | latest |
| QD2 | ECU_Identifier_Specification | latest |
| QD3 | Standard_Component | latest |
| QD4 | basic_diagnostic_System_specification | latest |
| QD5 | Flash_Specification | latest |

| | | |
|---|---|---|
| QD6 | Coding_Specification | latest |
| QD7 | Basic_DID_Definition_Specification | latest |
| QD8 | Part 2 Body Control Module (BCM) | latest |
| QD9 | System_schematics_Vinfast | latest |
| QD10 | CES_General_Electric | latest |
| QD11 | Part 2 Gateway (XGW) | latest |
| QD12 | Clamp Specification | latest |

*Table 30 Quoted documents*

## 5.2 Norms and Standards

| Ref. No. | Name | Version |
|---|---|---|
| NS1 | ISO 11898-1,<br>Road Vehicles -Controller area network (CAN) Part1<br>Data link layer and physical signalling. | 01.12.2003 |
| NS2 | ISO 11898-2, Road Vehicles - Controller Area Network (CAN) Part<br>2: High-speed medium access unit. | 01.12.2003 |
| NS3 | ISO 11898-3, Road Vehicles - Controller area network (CAN) Part<br>3: Low-speed, fault tolerant- medium dependent interface | 01.06.2006 |
| NS4 | AUTOSAR - Specification of Network Management Interface<br>Document Identification No: 228 Part<br>of Release: 4.3.1 | 08.12.2017 |
| NS5 | AUTOSAR – Specification of CAN Network Management<br>Document Identification No: 013 Part<br>of Release: 4.3.1 | 08.12.2017 |
| NS6 | ISO 15765-2 (2016-04-01), Road vehicles — Diagnostic communication over Controller Area Networks (DoCAN) — Part 2: Transport protocol and network layer services | 01.04.2016 |
| NS7 | ISO 15765-3, Road vehicles - Diagnostic on Controlled Area Networks (CAN) Part 3: Implementation of unified diagnostic services (UDS on CAN) | 2004(E),<br>15.10.2004 |
| NS8 | ISO/IEC 8859-1<br>Information technology -- 8-bit single-byte coded graphic character sets --<br>Part 1: Latin alphabet No. 1 | 1998 |
| NS9 | ISO 26262 Road vehicles – Functional safety | 01.08.2012 |
| NS10 | SAE J1850 Verification Test procedures | 05.06.2006 |

*Table 31 List of references*